# Waves & Sound

Subject: Physics | NoteShare Library

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WAVES & SOUND

1. WAVE EQUATION

2. TYPES
Transverse (light)
Longitudinal (sound)
Standing waves

3. SUPERPOSITION

4. BEATS

5. DOPPLER

6. SOUND
Speed in air ~ 343 m/s
Speed in water ~ 1500 m/s

7. STANDING WAVES
String: f_n = nv/(2L)
Open pipe: f_n = nv/(2L)
Closed pipe: f_n = (2n-1)v/(4L)

8. RESONANCE
Natural frequency